

Research Proposal on:
A Comprehensive Review and Hands-On Comparative Study of Few-Shot Learning Methods

Proposed for: First-Year Master Research Project Project nature: Comparative Review of Few Shot Learning Methods Supervisor: Pr. Hamza Khalfi
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Introduction

Few-shot learning is an important research direction in machine learning that aims to train models capable of learning new concepts from only a small number of labeled examples. This problem is highly relevant because many real-world domains suffer from data scarcity, high annotation cost, privacy constraints, or the need to rapidly adapt to new classes and tasks.

Instead of focusing only on a single algorithm, this project proposes a comprehensive review and practical comparison of multiple few-shot learning and meta-learning methods. The student will use the book repository Hands-On Meta Learning With Python as a practical guide for understanding and reproducing representative algorithms, including Siamese Networks, Prototypical Networks, Matching Networks, Relation Networks, Memory-Augmented Neural Networks, MAML, Meta-SGD, Reptile, and recent meta-learning variants.

The project is designed as a first-year master research project. Its goal is not necessarily to propose a new algorithm, but to build a structured understanding of the field, reproduce selected methods, compare their principles, strengths and limitations, and produce a clear technical report that can support future research in few-shot learning applications such as image classification, natural language processing, time series classification, or domain-specific recognition tasks.

Research Problem

The literature on few-shot learning contains many families of methods, including metric-based, optimization-based, memory-based, graph-based, and unsupervised meta-learning approaches. However, these methods are often presented in separate papers, with different datasets, evaluation protocols, implementation details, and terminology. This makes it difficult for a beginner researcher to understand the conceptual differences between methods and to identify which family of approaches is appropriate for a given problem.

The central research question of this project is:

How can the main few-shot learning methods be systematically reviewed, implemented, and compared from both theoretical and practical perspectives?

Objectives

1. Primary Objective: Prepare a comprehensive review and hands-on comparative study of representative few-shot learning methods.

2. Secondary Objectives:

- Study the foundations of few-shot learning, one-shot learning, meta-learning, support/query sets, and episodic training.
- Classify few-shot learning methods into meaningful families: metric-based, optimization-based, memory-based, graph-based, and unsupervised/task-generation methods.
- Learn With Python and Pytorch repositories to understand and reproduce selected algorithms.
- Compare the selected methods in terms of learning principle, architecture, adaptation mechanism, computational cost, implementation difficulty, and expected performance.
- Produce a reproducible set of notebooks or scripts illustrating at least three representative methods.
- Identify limitations, open challenges, and future research directions in few-shot learning.

Proposed Taxonomy of Methods

The review will organize few-shot learning methods according to the following taxonomy:

- Metric-based methods: Siamese Networks, Matching Networks, Prototypical Networks, Relation Networks.
- Optimization-based methods: MAML, First-Order MAML, Reptile, Meta-SGD.
- Memory-based methods: Neural Turing Machines and Memory-Augmented Neural Networks.
- Graph-based methods: Few-Shot Learning with Graph Neural Networks.
- Unsupervised and task-generation methods: CACTUs and related unsupervised meta-learning strategies.
- Recent perspectives: task-agnostic meta-learning, gradient agreement, cross-domain few-shot learning, and the role of strong pretrained embeddings.

Literature Review

1. Koch, G., Zemel, R., & Salakhutdinov, R. (2015). Siamese Neural Networks for One-shot Image Recognition. ICML Deep Learning Workshop.
2. Vinyals, O., Blundell, C., Lillicrap, T., Kavukcuoglu, K., & Wierstra, D. (2016). Matching Networks for One Shot Learning. Advances in Neural Information Processing Systems.
3. Santoro, A., Bartunov, S., Botvinick, M., Wierstra, D., & Lillicrap, T. (2016). Meta-Learning with Memory-Augmented Neural Networks. International Conference on Machine Learning.
4. Snell, J., Swersky, K., & Zemel, R. (2017). Prototypical Networks for Few-shot Learning. Advances in Neural Information Processing Systems.
5. Finn, C., Abbeel, P., & Levine, S. (2017). Model-Agnostic Meta-Learning for Fast Adaptation of Deep Networks. International Conference on Machine Learning.
6. Li, Z., Zhou, F., Chen, F., & Li, H. (2017). Meta-SGD: Learning to Learn Quickly for Few-Shot Learning. arXiv preprint arXiv:1707.09835.

7. Sung, F., Yang, Y., Zhang, L., Xiang, T., Torr, P. H. S., & Hospedales, T. M. (2018). Learning to Compare: Relation Network for Few-Shot Learning. IEEE/CVF Conference on Computer Vision and Pattern Recognition.
8. Nichol, A., Achiam, J., & Schulman, J. (2018). On First-Order Meta-Learning Algorithms. arXiv preprint arXiv:1803.02999.
9. Hsu, K., Levine, S., & Finn, C. (2018). Unsupervised Learning via Meta-Learning. arXiv preprint arXiv:1810.02334.
10. Garcia, V., & Bruna, J. (2018). Few-Shot Learning with Graph Neural Networks. International Conference on Learning Representations.
11. Chen, W.-Y., Liu, Y.-C., Kira, Z., Wang, Y.-C. F., & Huang, J.-B. (2019). A Closer Look at Few-shot Classification. International Conference on Learning Representations.
12. Hospedales, T., Antoniou, A., Micaelli, P., & Storkey, A. (2022). Meta-Learning in Neural Networks: A Survey. IEEE Transactions on Pattern Analysis and Machine Intelligence.
13. Ravichandiran, S. Hands-On Meta Learning With Python: Learning to Learn using One-Shot Learning, MAML, Reptile, Meta-SGD and more. GitHub repository: <https://github.com/sudharsan13296/Hands-On-Meta-Learning-With-Python>

Methodology

The project will be conducted in four phases. Each phase combines theoretical reading with practical implementation and documentation.

Phase 1: Context, Foundations and Familiarization

- Study the basic concepts of few-shot learning: N-way K-shot classification, support set, query set, episodic training, task distribution, and meta-training/meta-testing.
- Prepare a short technical note explaining the difference between classical supervised learning, transfer learning, and meta-learning.

Phase 2: Theoretical Review of Few-Shot Learning Families

- Study metric-based methods such as Siamese Networks, Matching Networks, Prototypical Networks, and Relation Networks.
- Study optimization-based methods such as MAML, First-Order MAML, Reptile, and Meta-SGD.
- Study memory-based and task-generation methods such as MANN and CACTUs.
- Build a comparative matrix describing each method, its key idea, mathematical principle, advantages, limitations, and typical use cases.

Phase 3: Practical Reproduction of Selected Methods

- Reproduce at least three representative methods from the repository: one metric-based method, one optimization-based method, and one advanced or memory-based method.
- Recommended minimum selection: Prototypical Networks, MAML or Reptile, and Siamese Networks or Relation Networks.
- Use small benchmark datasets such as Omniglot or mini-ImageNet

- Document all implementation issues, library compatibility problems, dataset preparation steps, and modifications required to run the code.

Phase 4: Comparative Analysis

- Compare the selected methods according to accuracy, training complexity, adaptation mechanism, number of parameters, implementation difficulty, and sensitivity to the number of shots.
- Analyze results under 1-shot and 5-shot settings whenever possible.
- Discuss the practical trade-off between simplicity and performance.
- Identify which methods are easier to reuse for future applications such as NER, time series classification, medical image classification, or domain-specific recognition.

Expected Deliverables

1. A comprehensive literature review report on few-shot learning and meta-learning methods.
2. A taxonomy of few-shot learning methods with a comparative table.
3. A reproducible implementation folder or notebook collection.
4. A practical reproduction report describing experiments, results, errors encountered, and code adaptations.
5. A comparative analysis of at least three representative methods.
6. A final presentation summarizing the taxonomy, experiments, findings, and future directions.
7. A short research paper-style synthesis that can later be expanded into a survey article or second-year master thesis proposal.

Expected Scientific Value

This project will help the student build a strong foundation in modern few-shot learning and meta-learning. It will also produce a useful pedagogical and technical resource for future students who want to work on low-data learning problems. The expected scientific value lies in the structured synthesis of several algorithmic families and in the reproducible comparison of representative methods.

The project can also serve as a preparation step for future applied research, for example few-shot named entity recognition, few-shot time series classification, few-shot medical diagnosis, or few-shot cybersecurity event detection.

Conclusion

This research project proposes a comprehensive theoretical and practical review of few-shot learning methods. By combining literature analysis with hands-on experimentation using Python and Pytorch code bases, the student will gain a deep understanding of how different meta-learning algorithms operate and how they compare in practice. The project is well suited for a first-year master student because it balances theoretical study, implementation, reproducibility, and critical analysis. It can also provide a strong foundation for a more specialized second-year master thesis in few-shot learning.

Appendix: Suggested Comparative Matrix

Method	Family	Main idea	Strength	Limitation
Siamese Network	Metric-based	Learns pairwise similarity between examples.	Simple and intuitive.	Pairwise training can be costly.
Matching Network	Metric-based	Classifies queries using attention over support examples.	Directly uses support set.	Sensitive to embedding quality.
Prototypical Network	Metric-based	Computes one prototype per class in embedding space.	Simple, strong baseline.	Prototype mean may fail for complex classes.
Relation Network	Metric-based	Learns a neural relation score between query and support examples.	Flexible learnable similarity.	Requires careful training.
MAML	Optimization-based	Learns an initialization that adapts quickly to new tasks.	General and model-agnostic.	Computationally expensive.
Meta-SGD	Optimization-based	Learns initialization, update direction, and learning rate.	Fast one-step adaptation.	More parameters to meta-learn.
Reptile	Optimization-based	First-order meta-learning by moving toward task-trained weights.	Simpler than MAML.	May require many episodes.

Practical Resource

The student should prepare his own GitHub repository as a practical starting point. If some other resource repositories are available, the student should list them, document them and, when necessary, adapt the implementations to his recently prepared repository using PyTorch environment.